DOCA Developer Day at NVIDIA GTC 2026

Build the Future of AI Infrastructure With NVIDIA DOCA [S81750]

March 17, 1:00 p.m. PT

Ariel Kit, Director, DOCA Products, NVIDIA

Yuval Degani, Sr. Director, Hyperscale Engineering, NVIDIA

Accelerated AI Networking With DOCA [S81857]

March 17, 2:00 p.m. PT

Alexander Petrovskiy, Sr. Product Manager, NVIDIA

Accelerate AI Inference Using DOCA for Storage [S81773]

March 17, 3:00 p.m. PT

Oren Duer, Principal Software Architect, NVIDIA

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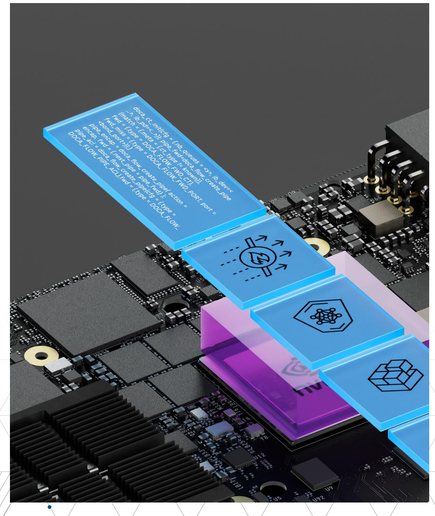
Omri Kahalon, Sr. Manager, Infrastructure Software Architect, NVIDIA

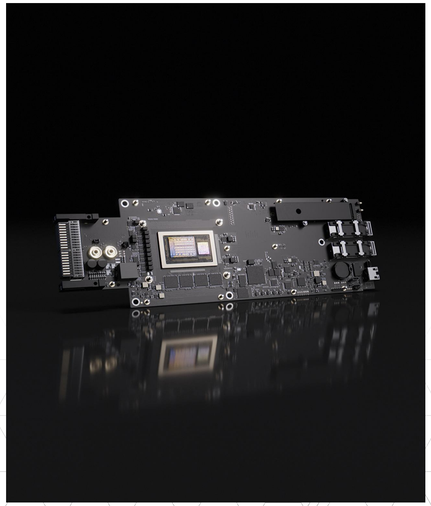
Build Secure AI Infrastructure With DOCA [S81625]

March 17, 4:00 p.m. PT

Ofir Arkin, Sr. Distinguished Architect, Cybersecurity, NVIDIA

Ariel Kit, Director, DOCA Products, NVIDIA





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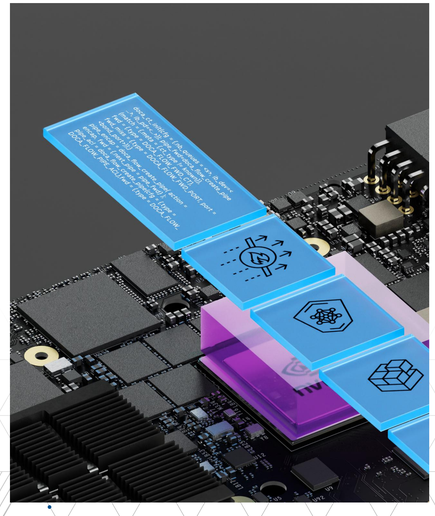

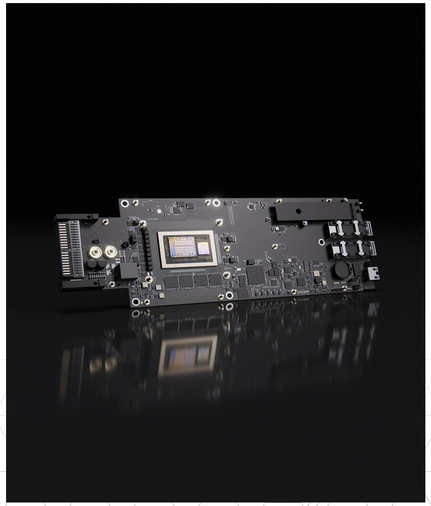

DOCA Developer Day

Accelerate AI Inference using DOCA for Storage

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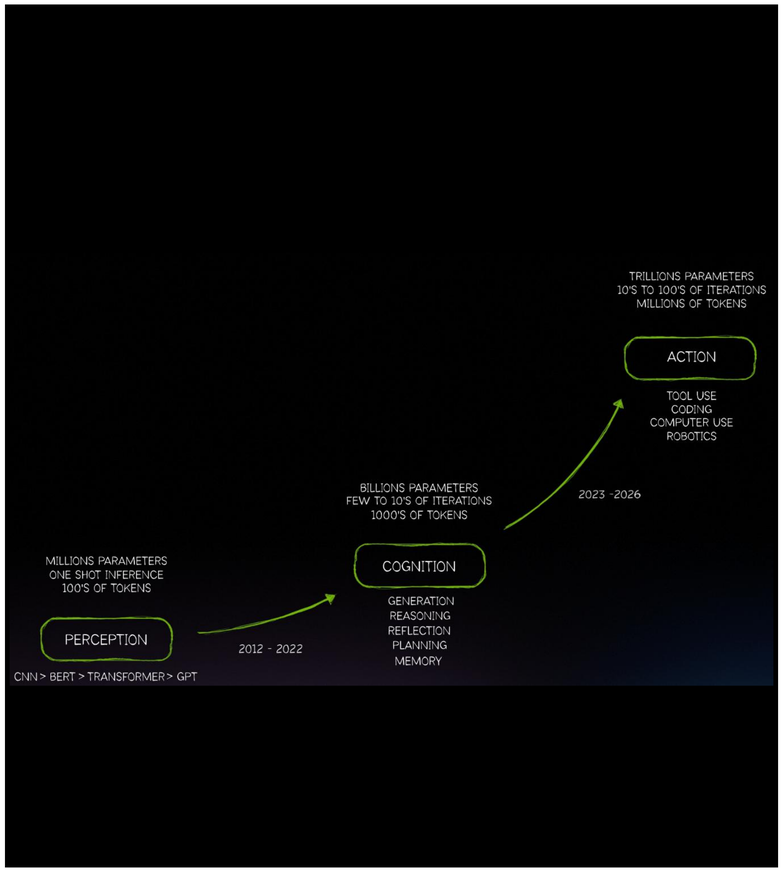
GTC 2026

Scaling AI Inference: The Agentic Shift

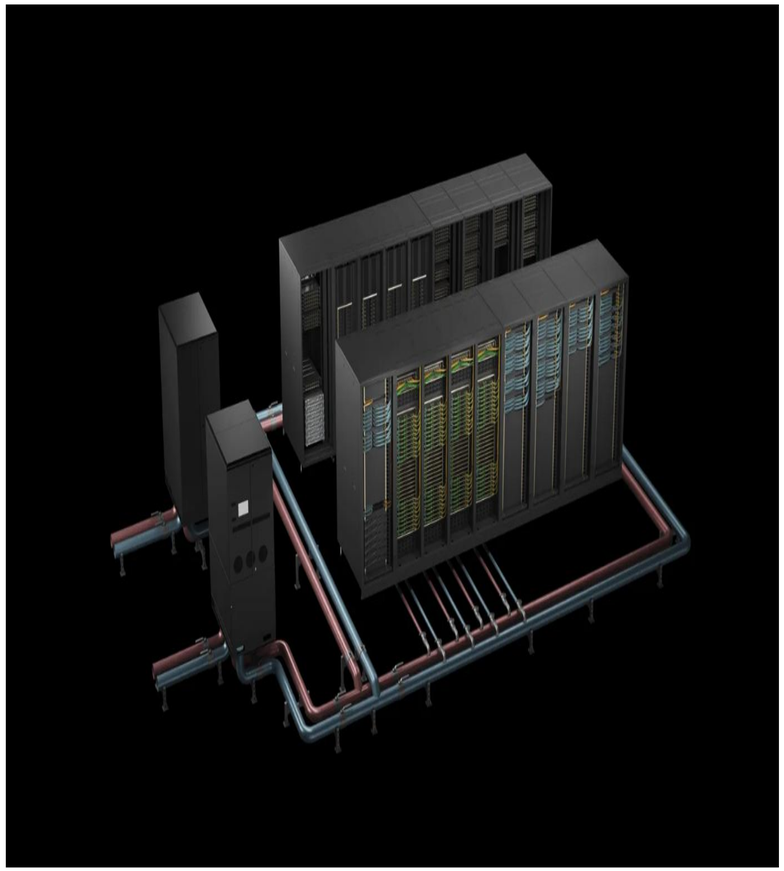
Agentic Workflows

Shift from response to result, using iterative reasoning and KV caching as a persistent working memory



Tokens Explosion

Agentic AI is increasing compute and token volume, Thinking tokens, Expanding context & sessions



The Power Ceiling

Energy efficiency the new scaling lever
Tokens per Watt is driving TCO

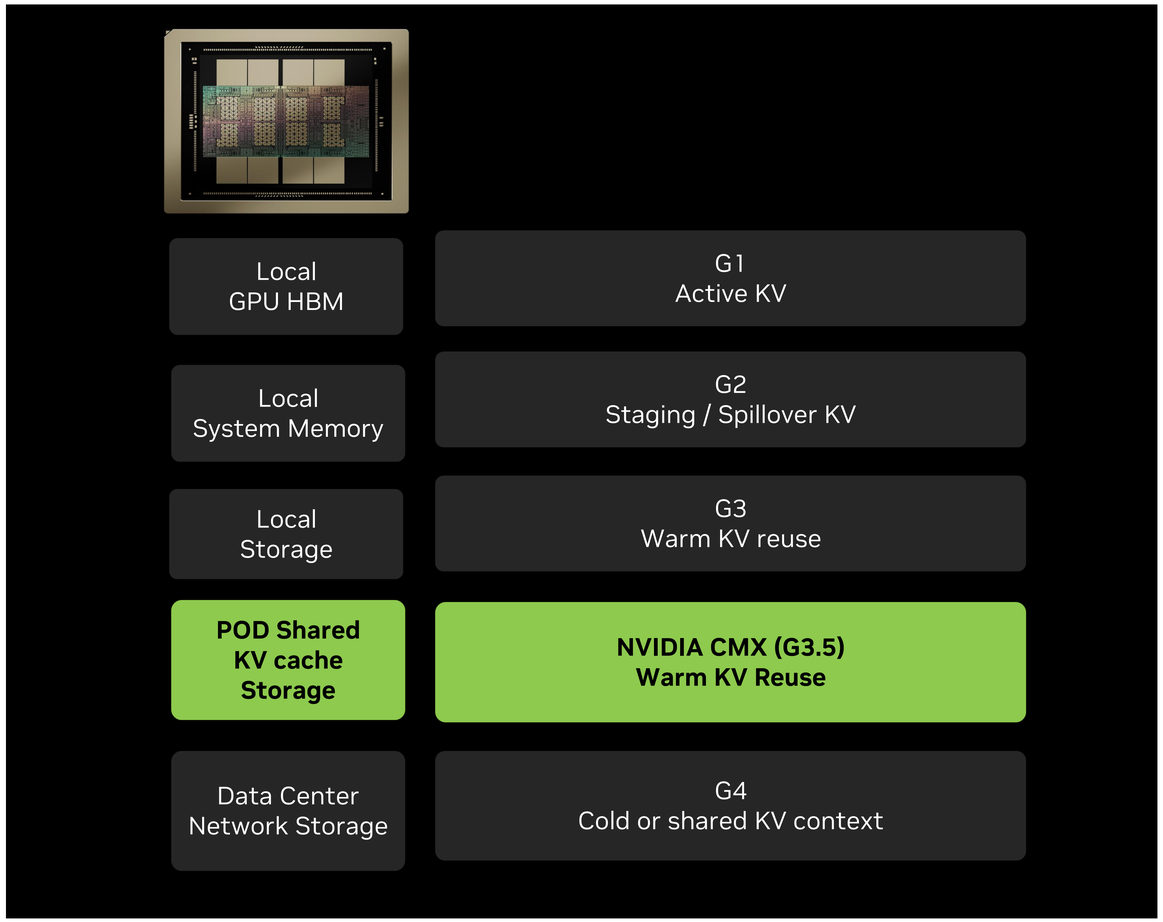
NVIDIA

Inference Context is the New Bottleneck

Introducing New Storage Tier, Purpose Built For KV Cache

- Inference context is large, immutable, and re-computable
- Performance depends on storing and reusing context
- System memory is inherently limited
- Scaling traditional storage increases cost and power
- Efficient context scaling demands a new architecture

- Need a new storage tier – G3.5, purpose built for KV cache



Context Memory Hierarchy

NVIDIA CMX Context Memory Storage Platform

NVIDIA BlueField powered storage infrastructure for the next frontier of AI

NVIDIA CMX Platform

- AI-native context storage tier
- Shared across GPUs for pod-level context reuse
- Complements GPU memory, local, and networked storage
- 5X higher token throughput and 5X higher power efficiency
- Accelerated by NVIDIA BlueField-4, Spectrum-X Ethernet, DOCA, and Dynamo
- Designed by NVIDIA and built by storage industry leaders

AIC CLOUDIAN ddn DELL Technologies HPE Hitachi Vantara IBM

NetApp NUTANIX PURESTORAGE SUPERMICRO VAST WEKA

NVIDIA DOCA 3.0 For Storage

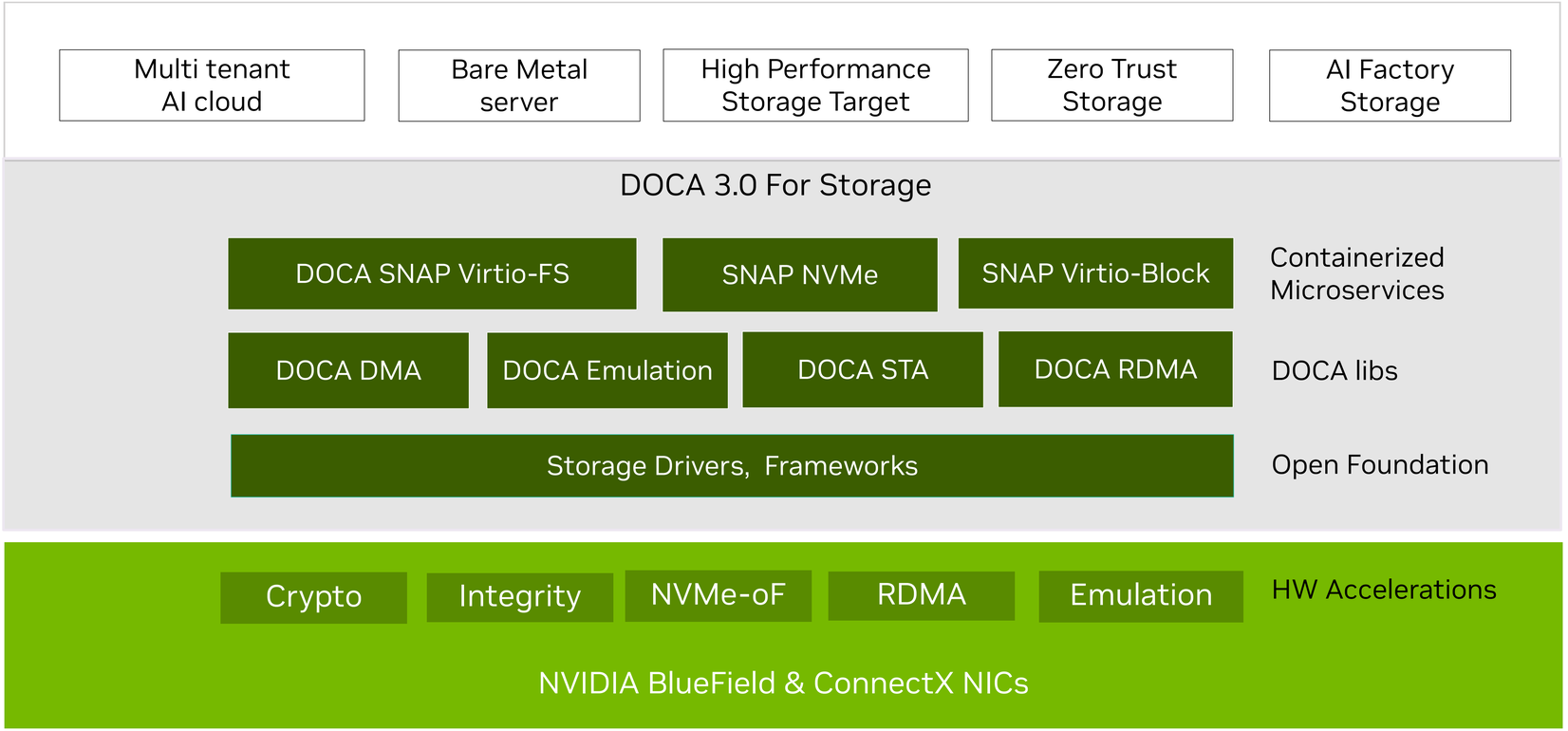
Powering Accelerated & Secured AI Storage Solutions

Disaggregated, virtualized storage at scale

HW accelerated data path

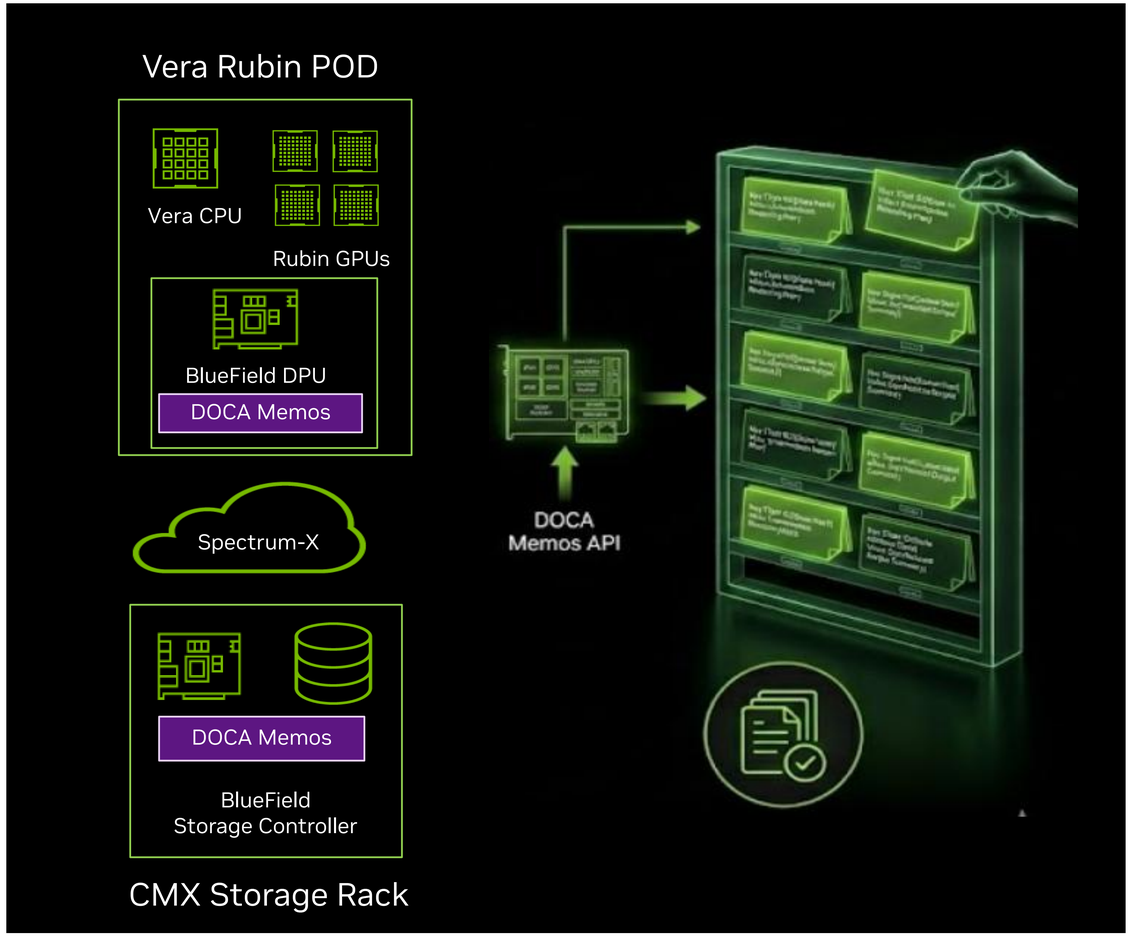
Secured, isolated, High performance storage access

Simple, flexible, operationally efficient storage infrastructure



Introducing DOCA Memos For The Agentic Era

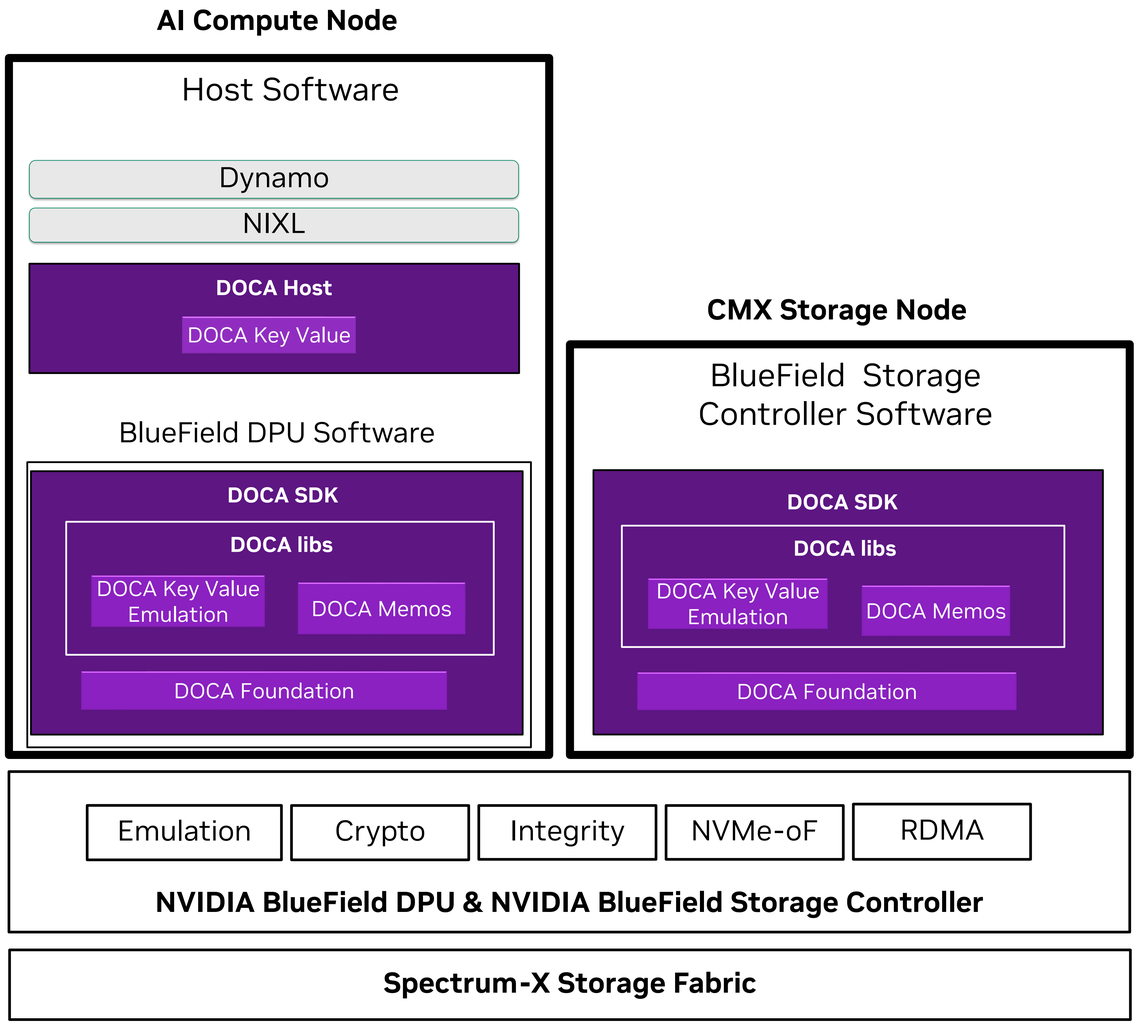
KV Cache Acceleration And Sharing At Pod Scale



- SDK for efficient KV cache data movement and sharing
- Runs on BlueField at compute and context memory nodes
- Purpose built for KV cache inference use case
- Key-value APIs, expose context directly to AI compute
- Developer-ready, extensible framework

DOCA Memos

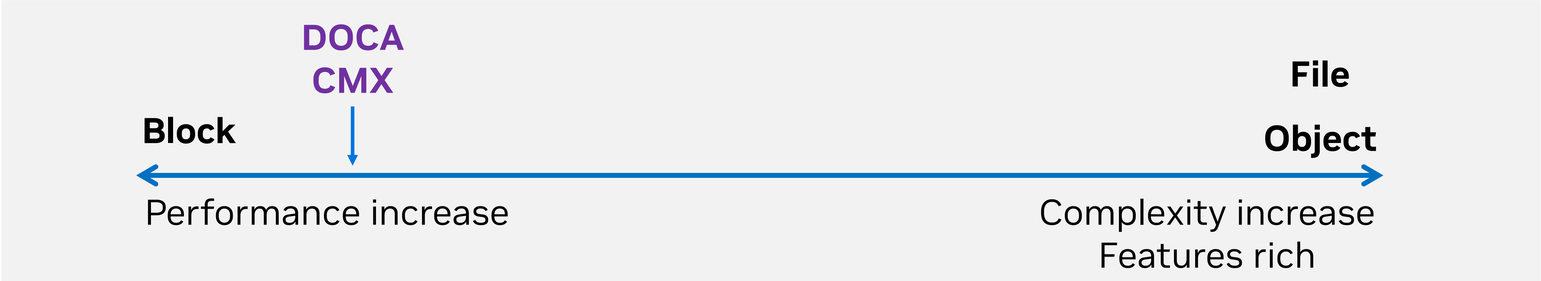
Unlocking Full Potential Of BlueField For CMX



- Tight integration with NVIDIA inference stack (Dynamo, NIXL)
- 800Gbps Spectrum-X, high effective bandwidth
- POD level scalable and power efficient
- Hardware-accelerated KV Cache integrity and encryption
- Elastic & secured KV cache volume
- Empower storage industry ecosystem

DOCA Memos – Purpose Built For KV Cache Use Case

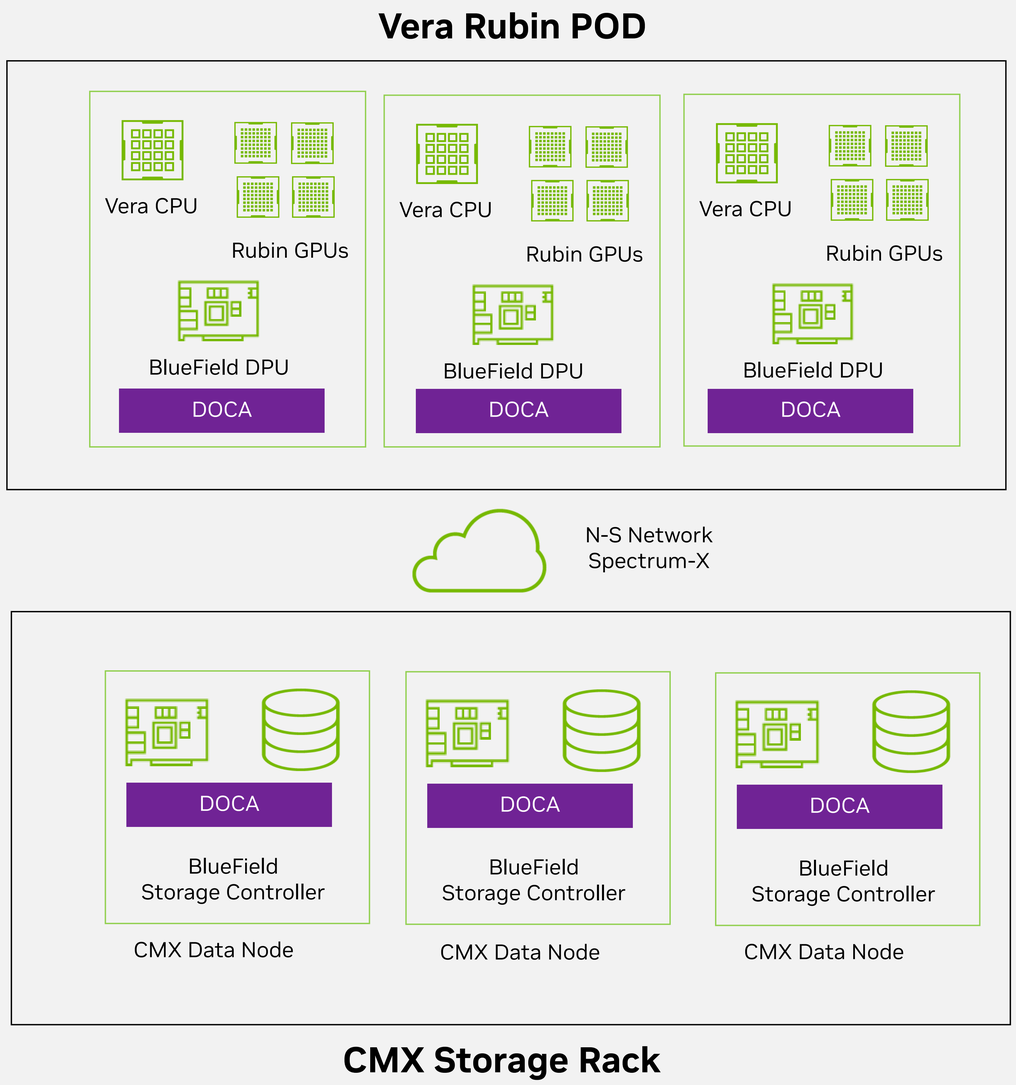
Enabling new storage class, leveraging unique properties of KV cache blocks and Agentic AI



Feature	DOCA Memos	Legacy file/object storage
I/O Size	Large (>1MB), fixed, pre-configured	Flexible, from 1B to GBs
Mutability	Immutable	Updates anywhere, at any granularity
Durability	<100%	100% guaranteed
Metadata	Minimal	Filenames, folders, keys, lookups, free/busy lists
Write Amplification	Close to 1	High
SSD Lifetime	Large IOs in favor of SSD durability	Must be general purpose

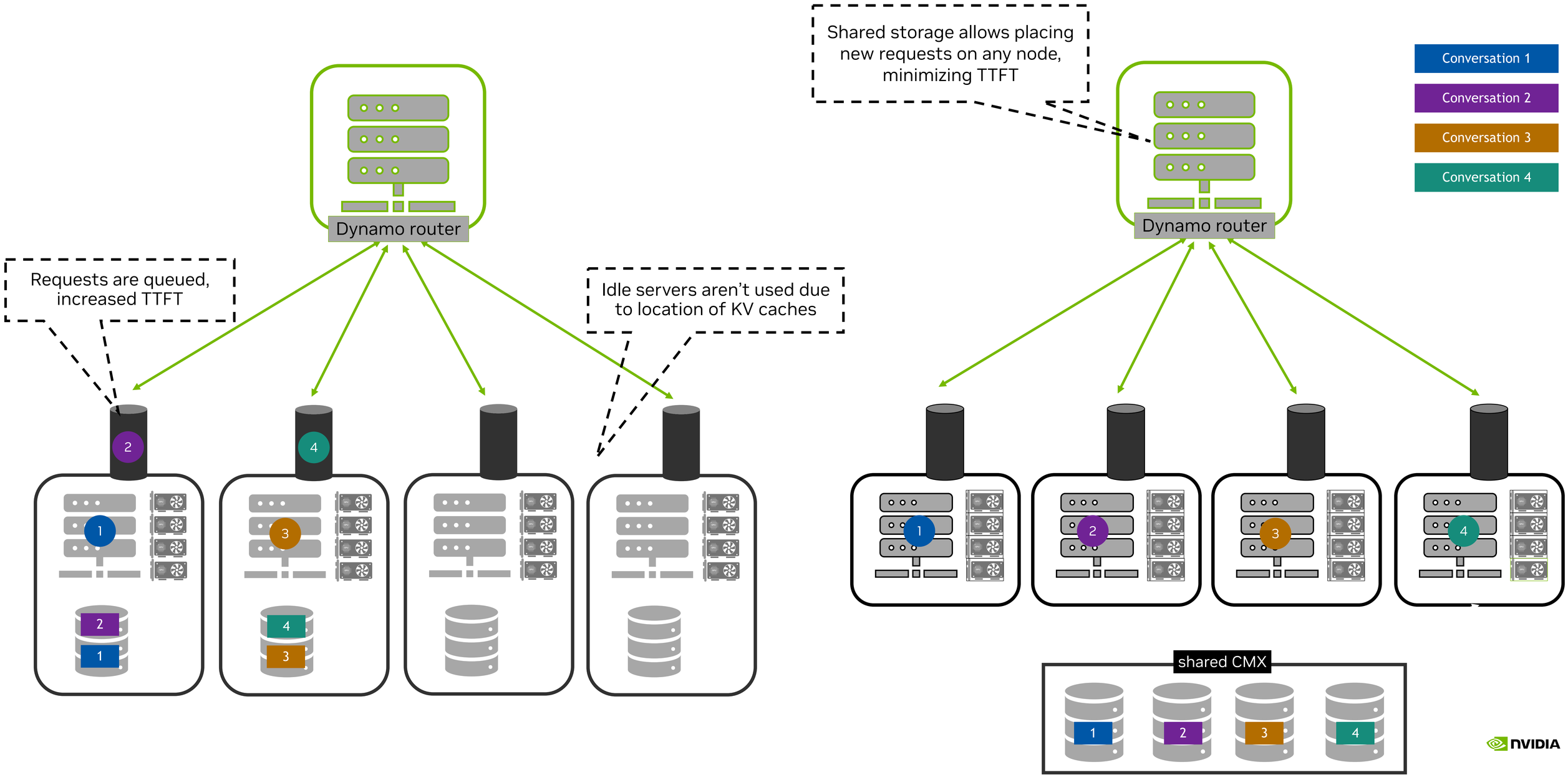
CMX Co-Design Advantages For Accelerated KV Cache System

Distributed KV Cache: End-to-End BlueField Roles

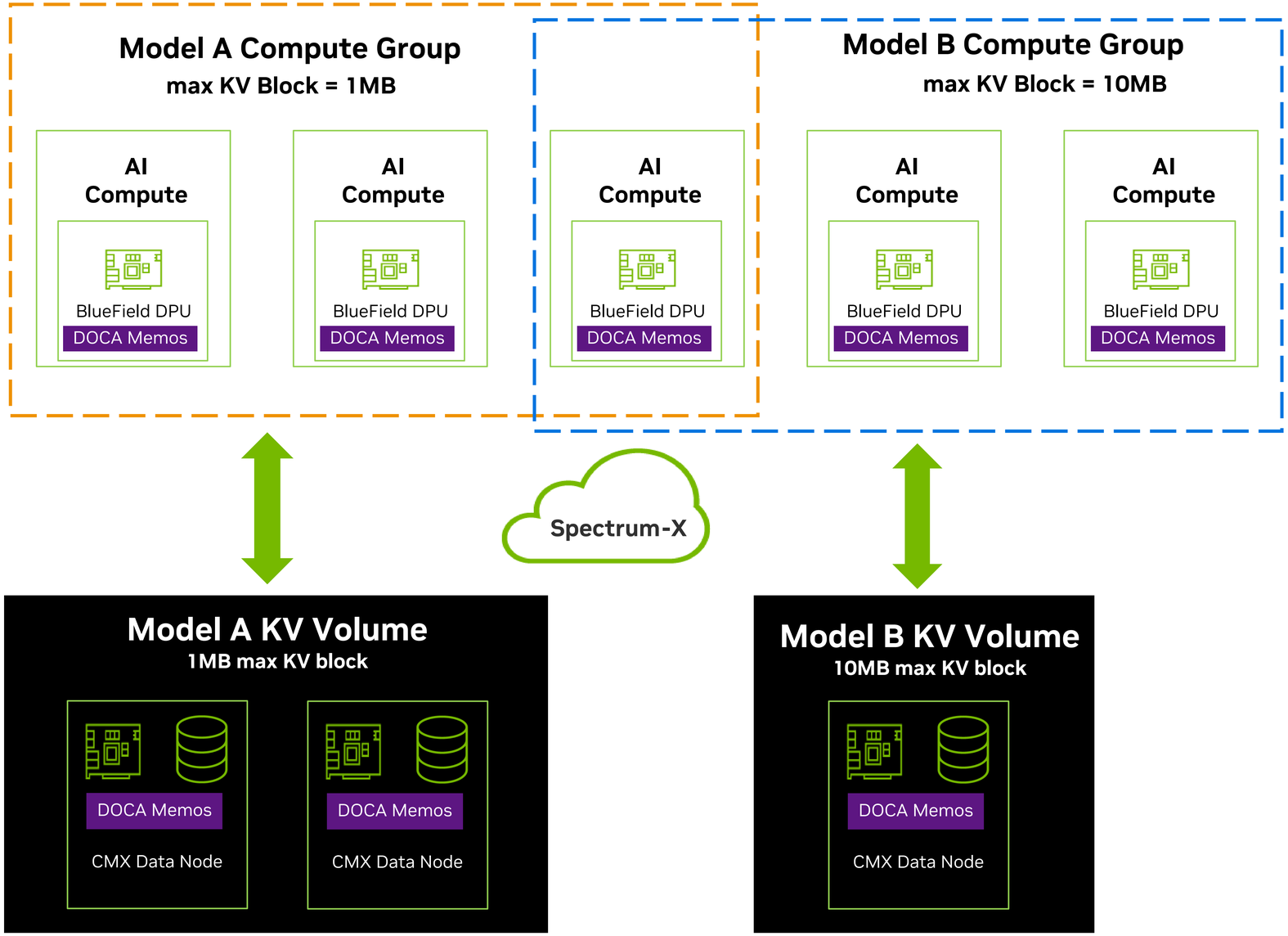


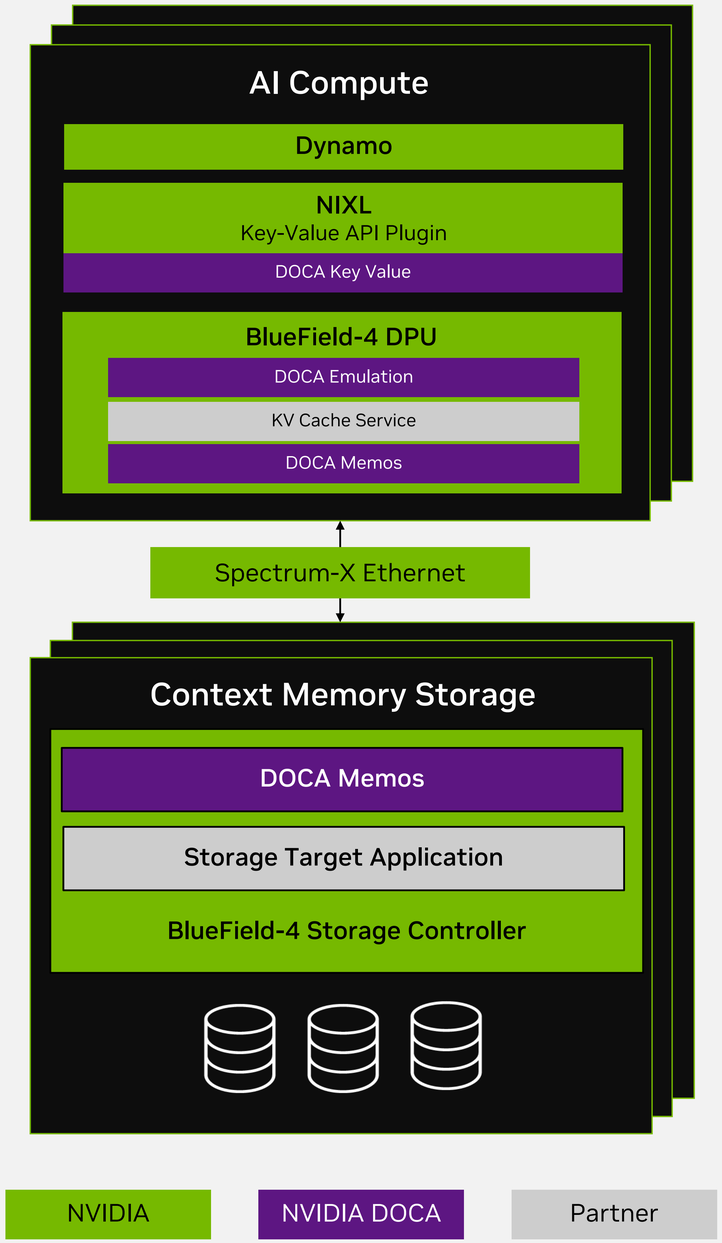
- KV cache pipeline is split for optimal latency, bandwidth, & sharing between
- **Compute-Side BlueField DPU: KV cache Mapping & Isolation**
  - Translates KV requests into KV-aware traffic with KV extensions & hints
  - Isolates KV logic and metadata execution environment with access control
- **CMX-Side BlueField Controller: Placement & Sharing**
  - Data placement on shared volumes with system awareness
  - Manage SSD data placement to maximize endurance & ensuring data consistency
  - Near data decisions for optimal data path utilization

DOCA Memos – Sharing Data Across GPUs In Block Storage Performance



CMX Storage Rack Management With DOCA Memos

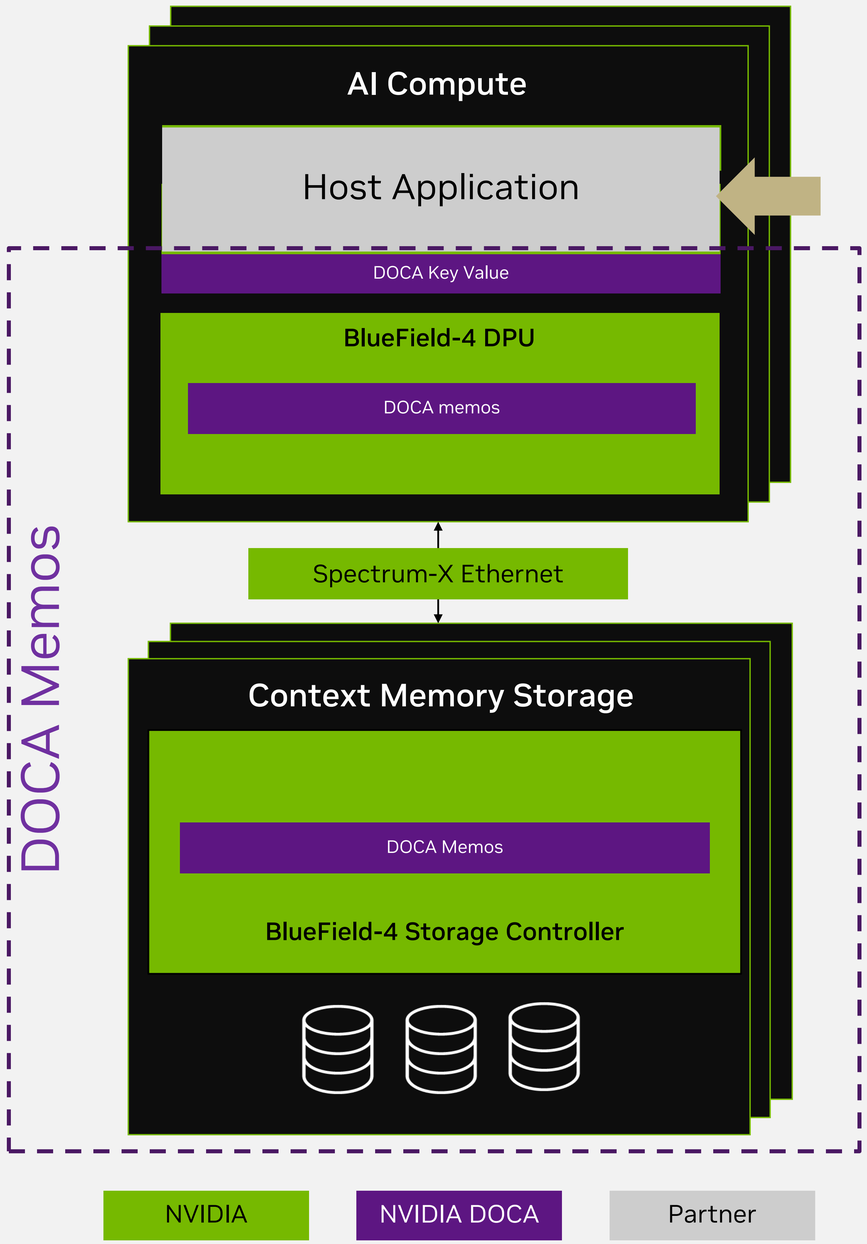




DOCA Memos Components

Secure, low-latency KV cache access at pod scale

- **Host**: DOCA Memos host driver
  - Provides C API for inferencing applications, generates KeyValue requests to DPU
    - `store(key, val)  // key=prefix_hash, val=KV`
    - `retrieve(key)`
- **DPU**: DOCA Memos initiator library
  - Provides C API for DPU emulation service, applies Memos logic, generates network requests to CMX
    - `store(key, val)  // key=prefix_hash, val=KV`
    - `retrieve(key)`
- **CMX:** DOCA Memos target library
  - Handles requests from network, applies additional Memos logic, provides C API for CMX Target software to read/write blocks from/to SSDs
    - `write_blocks_cb(data_ptr, disk, lba, len)`
    - `read_blocks_cb(data_prt, disk, lba, len)`



Host Applications Integrating With DOCA Memos

Requirements and Best Practices

- Up to 128 bit keys
- Don't gather per-KV cache metadata
  - Expecting "infinite" amount of KV cache blocks
  - Save to CMX with key=prefix_hash, retrieve later using same prefix_hash
- Use huge pages - to reduce amount of page translations
- Avoid `exists()` before `retrieve()`
  - `retrieve()` serves as lookup - fails fast if KV cache isn't found
- Correctly handle failed `retrieve()`
  - Successful `exists()` is not a guarantee...
  - Search in next tier or recalculate
- Support calculation of context "holes"
  - Re-calculate missing KV blocks in the middle, keep using the rest
- Modify workload routing to utilize shared storage benefits

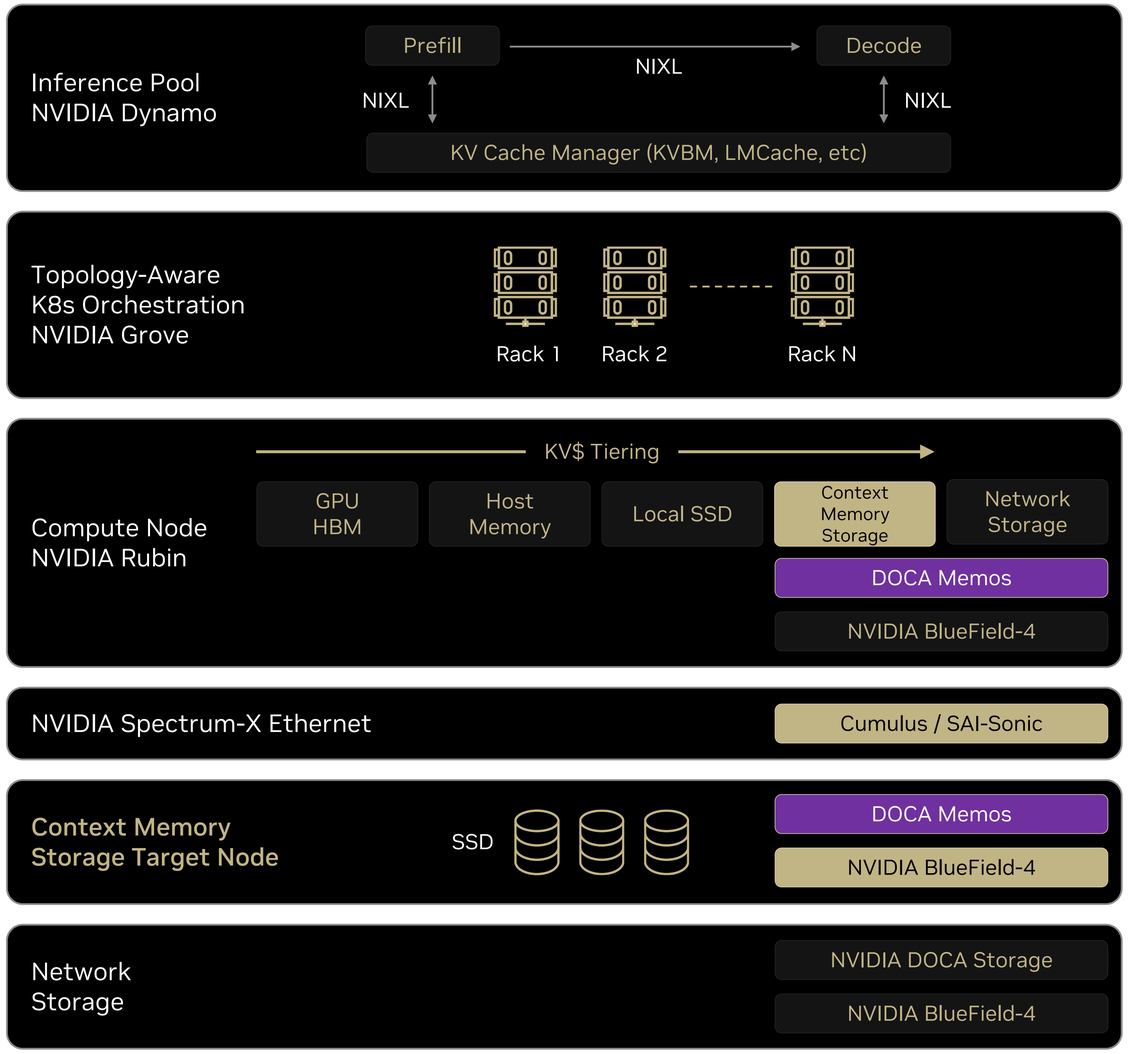
NVIDIA Context Memory Storage Platform

Pod-level architecture for sharing inference context at scale

KV Blocks Manager (Dynamo KVBM / LMCache)
- NIXL support for DOCA KV and128bit keys
- Huge pages support
- Avoid metadata tracking in host caching layer
- Support for fine grained error reporting for blocks

NIXL
- New NIXL *DOCA Memos* backend
  - Batched store/retrieve/exists support
- Fault tolerance and fine-grained error reporting

Future work
- Partial KV retrieval (“holes”)
- Metrics publishing by DOCA Memos
- Advanced KV eviction and offload filtering policies
- KV hints & workload aware signals

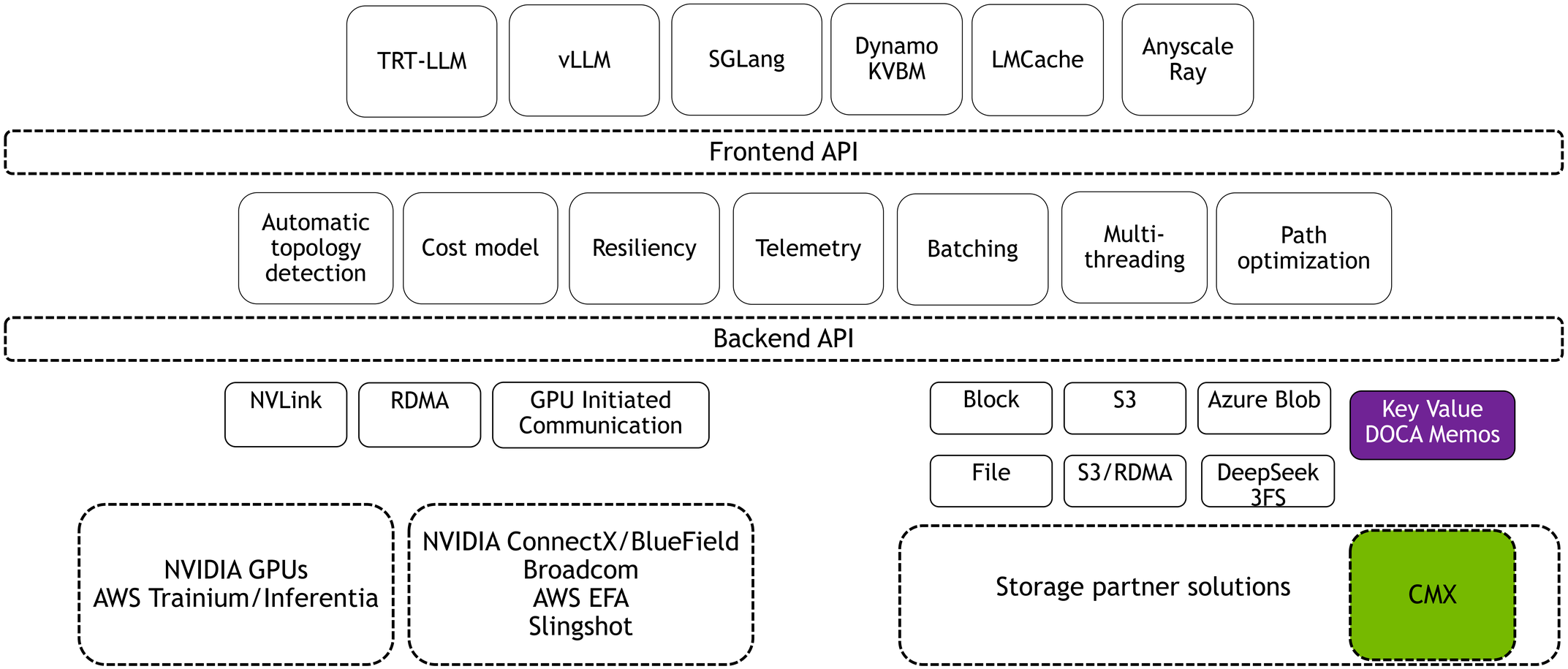


NVIDIA Inference Xfer Library (NIXL)

NIXL V1.0 is released!

pip install nixl

NIXL is a cross-vendor communication framework optimized for high-performance AI inference data transfers

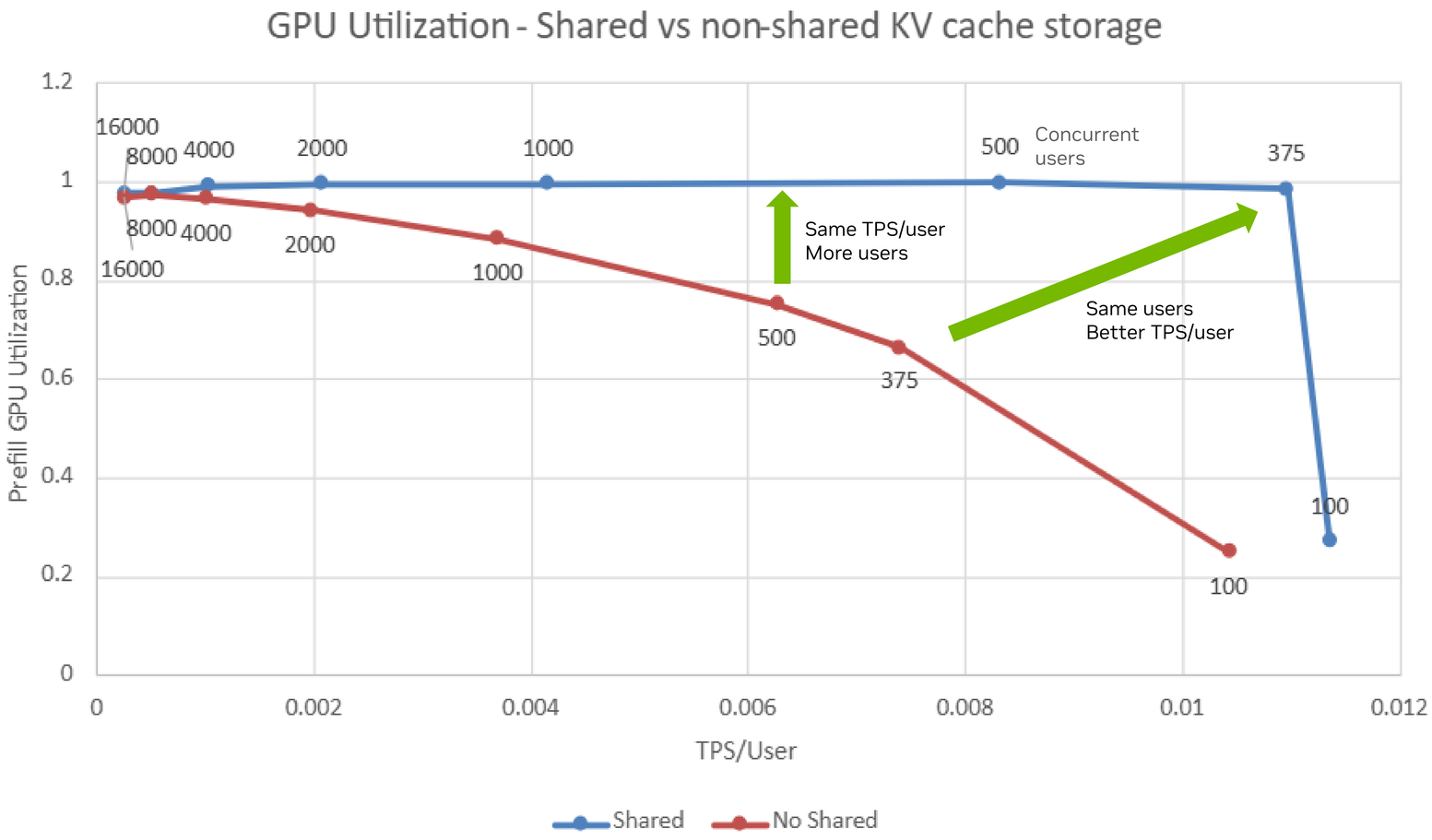


Simulation

Sharing vs. no sharing

- Simulation parameters
  - 100 GPUs
  - 64 steps per agent
  - 4K tokens added per step as new ISL
  - 128K cached tokens

- Takeaways
  - Sharing KV cache blocks allows 100% GPU utilization across larger range of TPS/User
  - Leads to:
    - Keep specific TPS/User for more users, or,
    - Improve TPS/User for the same amount of users

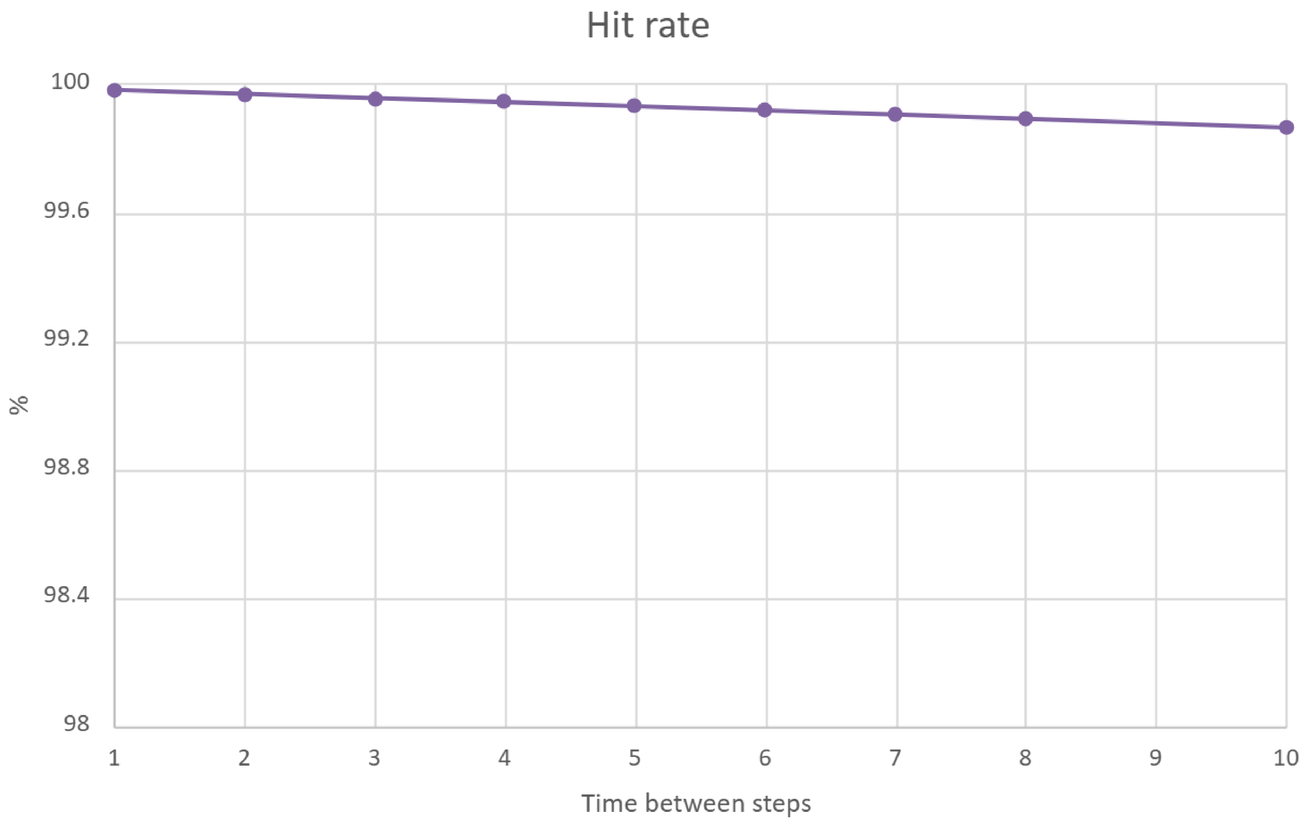


Simulation

DOCA Memos Hit Rate

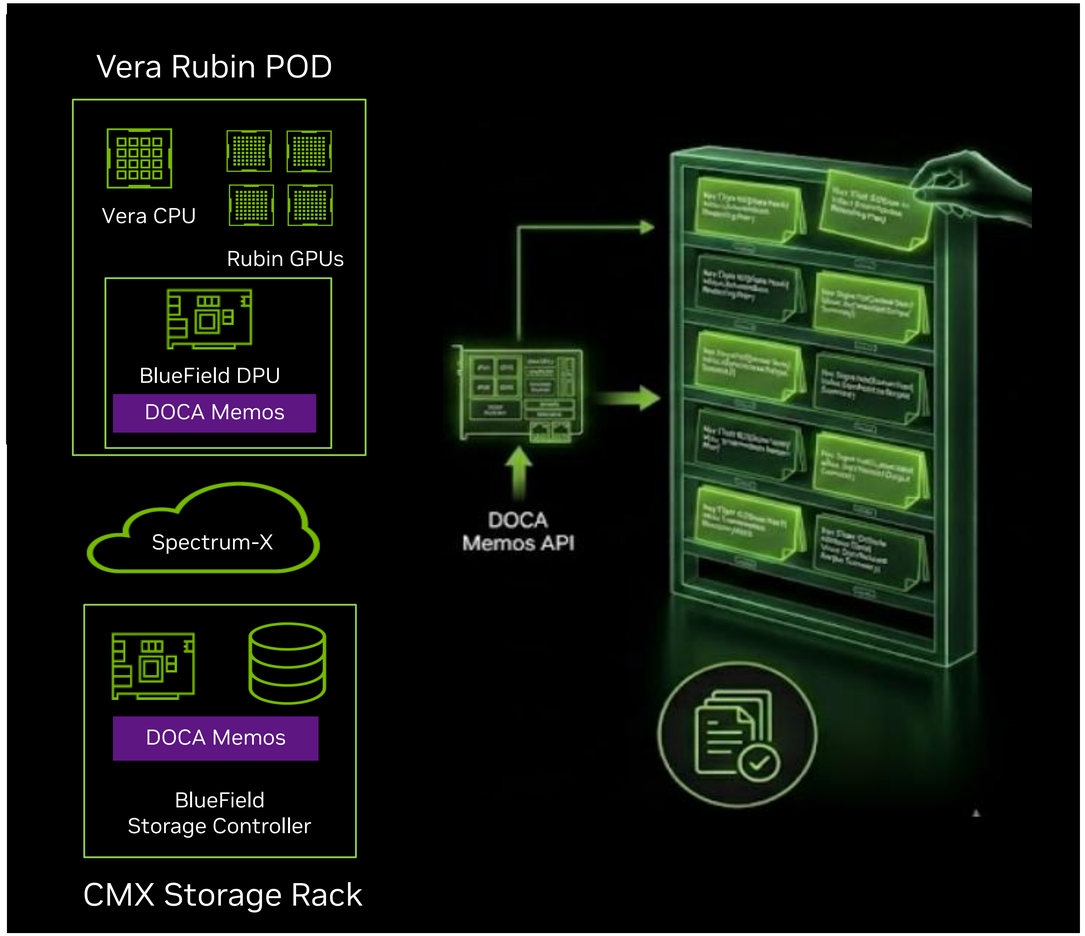
- Simulation parameters
  - GPT-OSS-20B, 1000 GPUs
  - Each agent 50 steps, 2K new tokens
    - Time in GPU 0.5s
  - 18 PB CMX storage
  - 700-5000 concurrent agents

- Takeaways
  - Hit rate is above 99.8% with planned CMX
  - GPU “goodput” is above 96%



DOCA Memos – Accelerated KV Cache At Pod Scale

Unlocking BlueField Potential For CMX



- ✓ The “agentic shift” – more KV needs faster & larger storage
- ✓ Scaling existing storage requires too much power & cost
- ✓ **CMX and DOCA Memos** provides new KV-specific storage solution
- ✓ **Dynamo, NIXL** implements necessary changes needed in inferencing systems
- ✓ Enables more users, longer context windows, larger models

NVIDIA